

	$f(t)$	$F(s)$
1	Unit Impulse $\delta(t)$	1
2	Unit Step $1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	t^n	$\frac{n!}{s^{n+1}}$
5	e^{-at}	$\frac{1}{s+a}$
6	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
7	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
8	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
9	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$
10	$\mathcal{L}[Af(t)]$	$AF(s)$
11	$\mathcal{L}[f_1(t) \pm f_2(t)]$	$F_1(s) \pm F_2(s)$
12	$\mathcal{L}_{\pm} \left[\frac{d}{dt} f(t) \right]$	$sF(s) - f(0\pm)$
13	$\mathcal{L}_{\pm} \left[\frac{d^2}{dt^2} f(t) \right]$	$s^2 F(s) - sf(0\pm) - \dot{f}(0\pm)$
14	$\mathcal{L}_{\pm} \left[\int f(t) dt \right]$	$\frac{F(s)}{s}$
15	$\mathcal{L}_{\pm} [f(t-a)1(t-a)]$	$e^{-sa} F(s)$
16	$e^{-at} f(t)$	$F(s+a)$